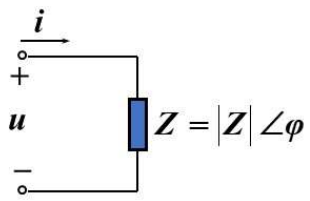
习题课3

- 正弦稳态电路

  相量法，复数阻抗，复数导纳
- 第三-五章作业讲评

第3次习题课

1.



电路如图所示，已知电压和电流为

$$u(t) = 10\sin(400\pi\, t + 60^\circ)\ \text{V}$$

$$i(t) = -\frac{1}{\sqrt{2}}\cos(400\pi\, t - 150^\circ)\ \text{A}$$

(a) $\omega =$ 400π rad/s, $f =$ 200Hz, $T =$ 0.005s .

(b) 有效值 $U =$ 7.07V，有效值 $I =$ 0.5A .

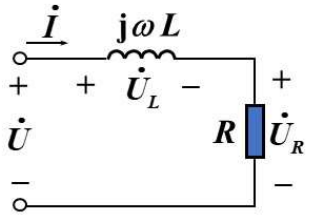
(c) u 和 i 的相位差 $\psi_u - \psi_i =$ -60° .

(d)负载是 容性，$|Z| =$ 14.14Ω，$\varphi =$ -60° .

单选题 1分

(3) $U = U_L + U_R$

这个表达式正确吗?



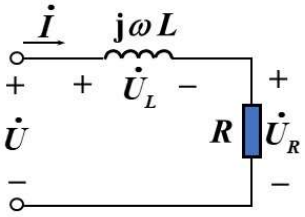
A 正确

B 不正确

单选题 1分

(5) $Z = \sqrt{R^2 + (\omega L)^2}$

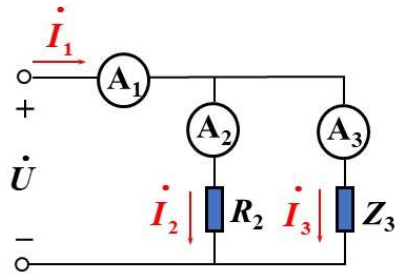
这个表达式正确吗?



A 正确

B 不正确

4.



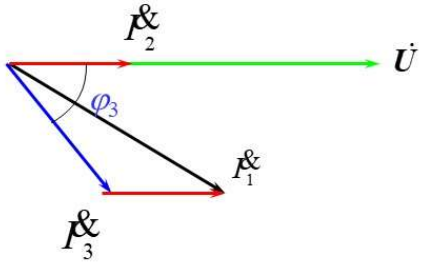
U=220V，f=50Hz，A_1 的读数是 4A，A_2 的读数是2A，A_3的读数是 3A，Z_3 是感性的．求 R_2 和Z_3。

$$R_2 = 110\ \Omega$$

$$Z_3 = \frac{220\angle 0^\circ}{3\angle -75.5^\circ}$$

$$Z_3 = 18.4 + j71\ \Omega$$

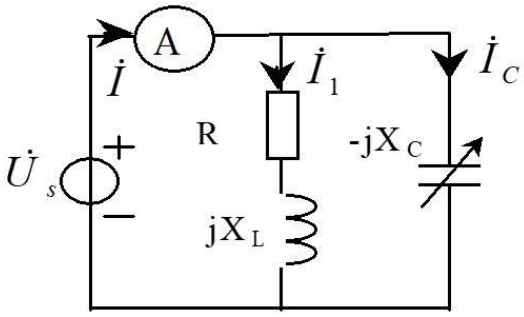
设 $\dot{U} = 220\angle 0^\circ \text{V}$

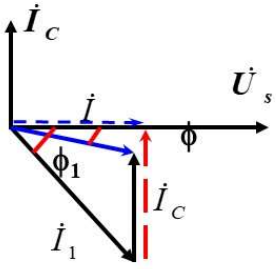


$$4^2 = 3^2 + 2^2 - 2\times 3\times 2\times \cos(180^\circ - \varphi_3)$$

$$\varphi_3 = 75.5^\circ$$

Ex：Us=380V,f=50Hz,电容C可调，当电容C=80.95μF时，电流表的读数最小，为2.59A.求I_1=?.





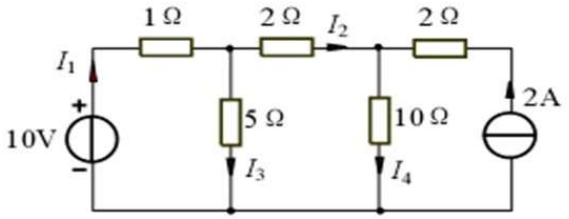
$$\dot{I}_C = j\omega C\dot{U}_s \Rightarrow I_C = 100\times 3.14\times 80.95\times 10^{-6}\times 380$$

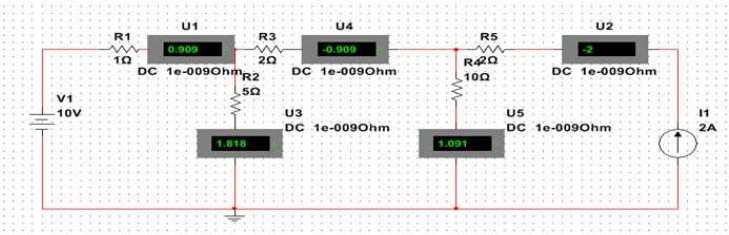
$$I_1 = \sqrt{I_C^2 + I^2}$$

2023年4月19日

3-1、用支路电流法求图所示电路中各支路电流及各电阻上吸收的功率。

解





R_1的功率 $P_1 = \frac{100}{121} = 0.826\ W$

$P_2 = \frac{2000}{121} = 16.528\ W$

$P_3 = \frac{200}{121} = 1.653\ W$

$P_4 = \frac{1440}{121} = 11.903\ W$

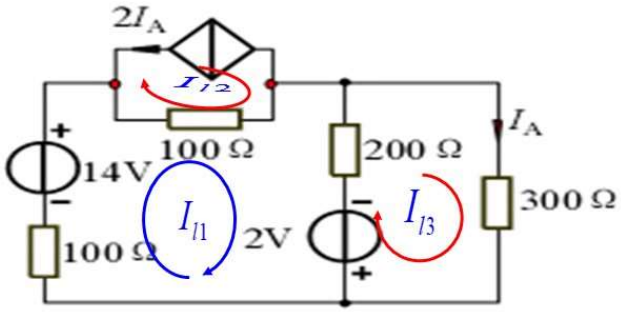
$P_5 = 8\ W$

电压源功率 $Pv = -\frac{100}{11} - 9.091\ W$

电流源功率 $Pi = -\frac{328}{11} = -29.818\ W$

3-2、 用网孔电流分析法求图示电路的各支路电流。

解



$$400\ I_{l1} - 100\ I_{l2} - 200\ I_{l3} = 16$$

$$I_{l2} = -2I_A$$

$$-200I_{l1} + (200 + 300)I_{l3} = -2$$

$$I_{l3} = I_A$$

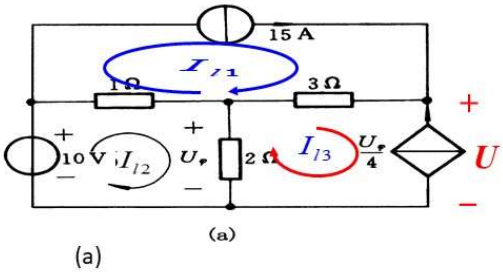
$I_{l1} = 0.04\text{A}$

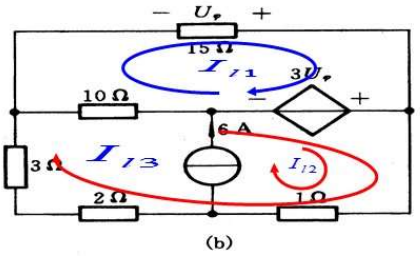
$I_{l1} = -0.024\text{A}$

$I_{l3} = 0.012\text{A}$

$I_A = 0.012\ A$

3-3、图示电路，试用回路电流法求受控源的功率。





解

(a)

$$I_{l1} = 15$$

$$-I_{l1} + (1+2)I_{l2} - 2I_{l3} = 10$$

$$I_{l3} = -\frac{U_\varphi}{4}$$

$$U_\varphi = 2(I_{l2} - I_{l3})$$

(b)

$$U_\varphi = 20\,\mathrm{V}$$

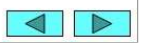
$$I_{l2} = 5\,\mathrm{A}$$

$$I_{l3} = -5\,\mathrm{A}$$

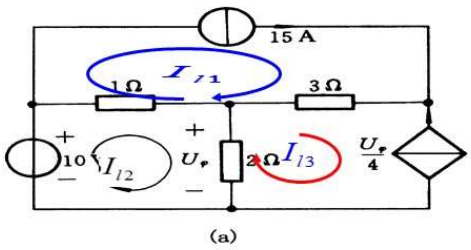
$$2(I_{l3} - I_{l2}) + 3(I_{l3} - I_{l1}) = -U$$

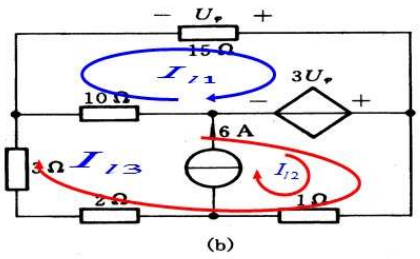
$$U = 80\,V$$

$$P_{发} = UU_\varphi / 4 = 400\,\mathrm{W}$$



3-3、图示电路，试用回路电流法求受控源的功率。





解

(b)

$$25I_{l1} - 10I_{l3} = -3U_\varphi$$

$$I_{l2} = 6$$

$$-10I_{l1} + I_{l2} + (10+1+3+2)I_{l3} = 3U_\varphi$$

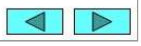
$$U_\varphi = -15I_{l1}$$

$$U_\varphi = 30\,\mathrm{V}$$

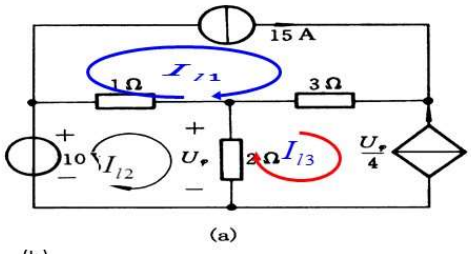
$$I_{l1} = -2\,\mathrm{A}$$

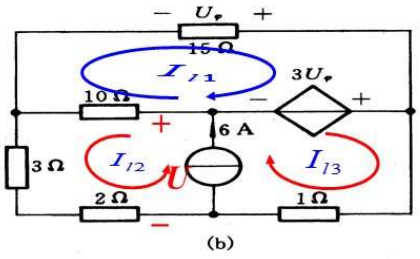
$$I_{l3} = 4\,\mathrm{A}$$

$$P_{吸} = 3U_\varphi(I_{l1} - I_{l2} - I_{l3}) = -1080\,\mathrm{W}\,(发\ 出)$$



3-3、图示电路，试用回路电流法求受控源的功率。





解法2

(b)

$$(15+10)I_{l1}+10I_{l2}=-3U_{\varphi}$$

$$(10+3+2)I_{l2}+10I_{l1}=-U$$

$$I_{l3}+U=3U_{\varphi}$$

$$U_{\varphi}=-15I_{l1}$$

$$I_{l2}+I_{l3}=6$$

$$I_{l1}=-2\ A$$

$$I_{l2}=-4\ A$$

$$I_{l3}=10\ A$$

$$U_{\varphi}=30\ V$$

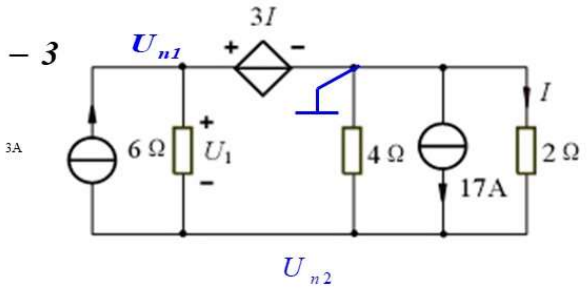
$$U=80\ V$$

$$P=-3U_{\varphi}(I_{l3}-I_{l1})=-1080\ W$$

3-4、试用结点法求图示电路中的U_1

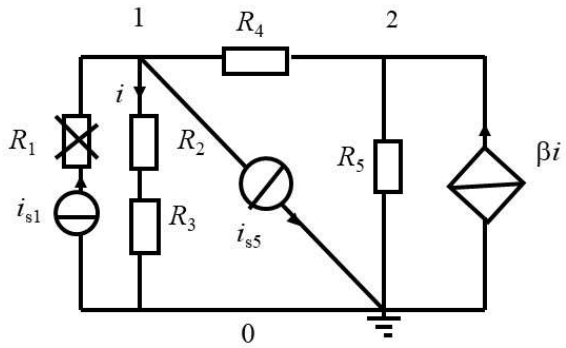
解

$$\begin{cases} U_{n1}=3I \\ -\frac{1}{6}U_{n1}+(\frac{1}{6}+\frac{1}{4}+\frac{1}{2})U_{n2}=17-3 \\ I=-\frac{1}{2}U_{n2} \\ U_1=U_{n1}-U_{n2} \end{cases}$$



$$U_1=-30\ V$$

3-5



$$(\frac{1}{R_2+R_3}+\frac{1}{R_4})u_{n1}-\frac{1}{R_4}u_{n2}=i_{s1}-i_{s5}$$

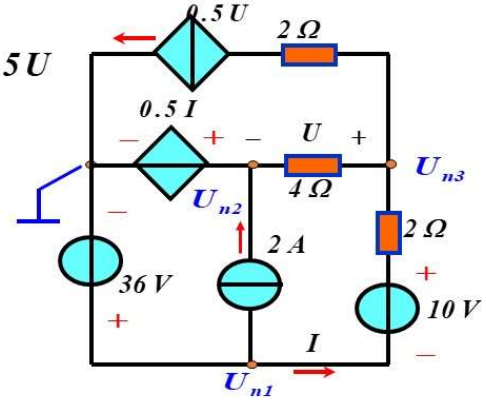
$$(\frac{1}{R_5}+\frac{1}{R_4})u_{n2}-\frac{1}{R_4}u_{n1}=\beta i$$

$$\frac{u_{n1}}{R_2+R_3}=i$$

3-6、 如图所示电路，求受控电流源输出的功率。

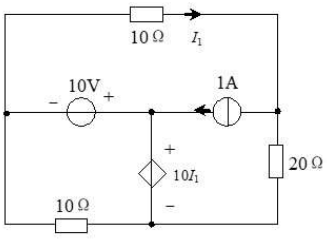
解

$$\begin{cases} U_{n1}=36 \\ U_{n2}=0.5I \\ -\frac{1}{2}U_{n1}-\frac{1}{4}U_{n2}+(\frac{1}{4}+\frac{1}{2})U_{n3}=5-0.5U \\ I=\frac{U_{n1}-U_{n3}+10}{2} \\ U=U_{n3}-U_{n2} \\ P_{0.5U}=(U_{n3}-2\times 0.5U)\times 0.5U(\text{吸收}) \end{cases}$$



$$\Rightarrow P_{0.5U}=48W(\text{吸收})$$

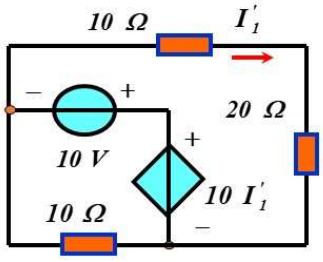
4-1、试用叠加定理求图示电路中的电流I_1



解　1、10V电压源单独作用时：

$$-10+10I_1'-30I_1'=0$$

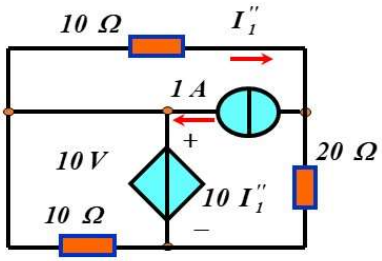
$$\Rightarrow I_1'=-0.5\,A$$



2、1A电流源单独作用时：

$$10I_1''+20(I_1''-1)-10I_1''=0$$

$$\Rightarrow I_1''=1\,A$$



$$I_1=I_1'+I_1''=0.5\,A$$

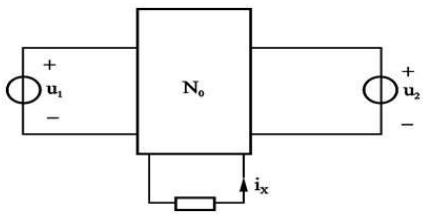
4-2、 图示电路中N_O为无源电阻网络。当U_1=1V，U_2=2V时；I_X=12A；又当U_1=-1V，U_2=2V时，I_X=0A。若将N_O变换为含有独立电源的网络后，在$U_1=U_2$=1V，时，I_X=-1A，试求当$U_1=U_2$=3V时的电路I_X。

解　1、N_O为无源电阻网络时：

由叠加定理：$I_X=K_1U_1+K_2U_2$

由已知条件：

$$\begin{cases}12=K_1\cdot 1+K_2\cdot 2\\0=K_1(-1)+K_2\cdot 2\end{cases}$$



2、N_O为含有独立电源的网络时：　内部电源产生的分量记为 I'''

$$I_X=K_1U_1+K_2U_2+I'''$$

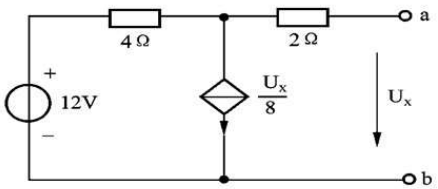
由已知条件：$-1=K_1\cdot 1+K_2\cdot 1+I'''$

$$\Rightarrow I_X=K_1\cdot 3+K_2\cdot 3+I'''=17\,A$$

雨课堂 Rain Classroom

4-3.、试求图示(a)和(b)含源一端口网络ab的戴维宁和诺顿等效电路。

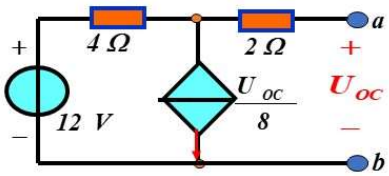
解　a、求U_{OC}：

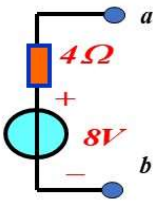


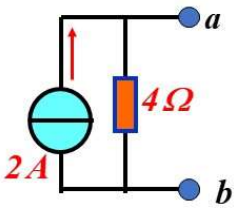
$$U_{oc} = -4\frac{U_{oc}}{8} + 12 \quad \Rightarrow \quad U_{oc} = 8V$$

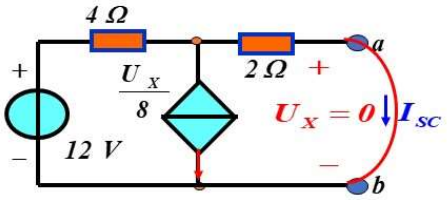
短路法求R_{eq}：　　短路时U_X=0：

$$I_{sc} = \frac{12}{4+2} = 2\,A \qquad R_{eq} = \frac{U_{oc}}{I_{sc}} = 4\,\Omega$$



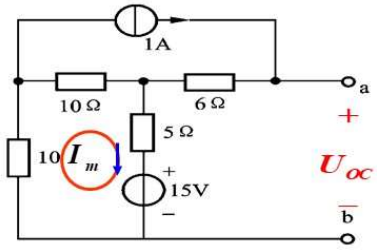


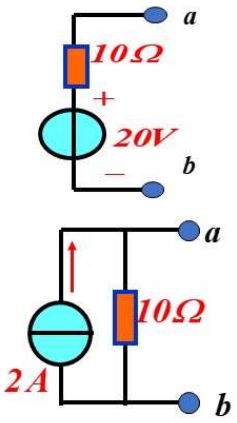




4-4、试求图示(a)和(b)含源一端口网络ab的戴维南和诺顿等效电路。

解　b、对回路：





$$25\,I_m - 10\cdot 1 = -15 \quad \Rightarrow \quad I_m = -0.2\,A$$

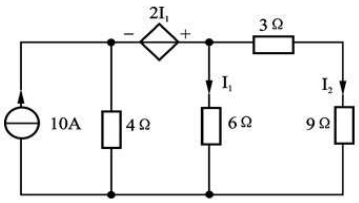
$$\therefore \quad U_{oc} = 6\cdot 1 + 5\,I_m + 15 = 20\,V$$

$$R_{eq} = 6 + 20 \,//\, 5 = 6 + \frac{20\cdot 5}{20+5} = 10\,\Omega$$

4-4、利用戴维宁定理或诺顿定理求图示电路中的电流I_2

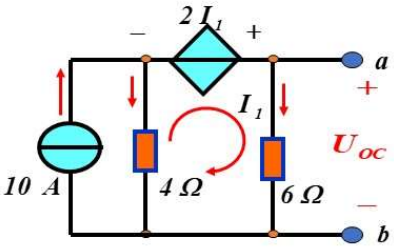
解　1、求U_{OC}：　$-4(10-I_1)-2I_1+6I_1=0$

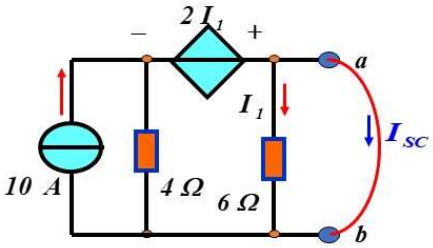
$\Rightarrow$　$U_{OC}=6I_1=30V$

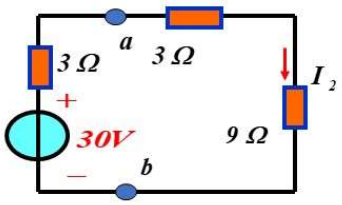


2、求R_{eq}：　短路法　短路时$I_1=0$：

$\Rightarrow$　$I_{SC}=10\ A$　$R_{eq}=\dfrac{U_{OC}}{I_{SC}}=3\Omega$　3、$I_2=\dfrac{U_{OC}}{3+3+9}=2\ A$

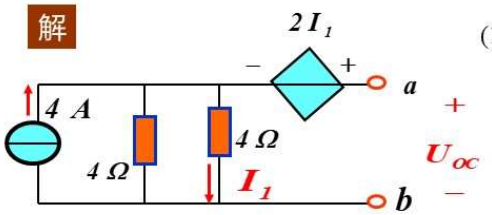


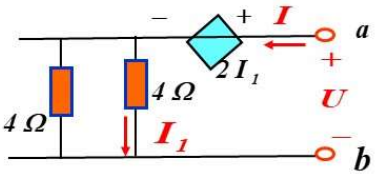




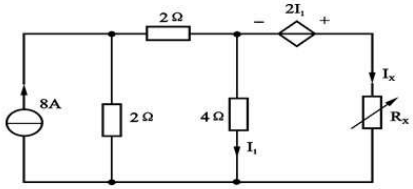
4-5、图示电路中的负载电阻R_X可变，，试问：(1)当R_X=6Ω时，电流I_X等于多少？（2）R_X等于多少时，可吸收最大功率？并求此功率。

解





(1) 求等效电路



求U_{OC}：

$I_1=2\ A$

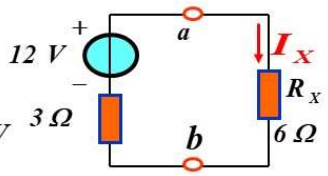
$U_{OC}=2I_1+4I_1=6I_1=12V$

求R_{eq}（外加电源法）　$I_1=\dfrac{1}{2}I$

$R_{eq}=\dfrac{U}{I}=\dfrac{2I_1+4I_1}{I}=3\Omega$

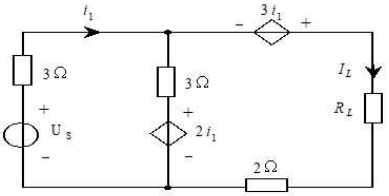
(2) 当R_X=6Ω时,求　$I_X=\dfrac{12}{3+6}=\dfrac{4}{3}A$

(3) 当R_X=3Ω时,有最大功率　$P_M=\dfrac{U_{OC}^2}{4R_X}=12\ W$

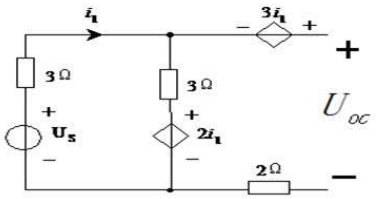


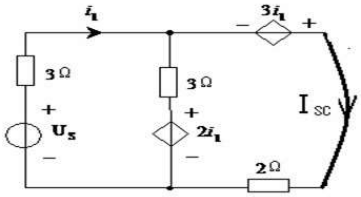
4-6、如图所示直流电路中，当电流I_L=2A时，负载电阻R_L消耗的功率最大，试求U_S。

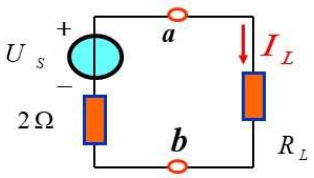
解　1、求U_{OC}：　　$\because I_L = 0$



$$U_{OC} = U_S - 3I_1 + 3I_1 - 2\times 0 = U_S$$







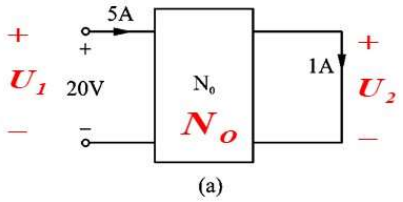
2、求R_{eq}：　　短路法　　$U_S - 3I_1 + 3I_1 - 2\times I_{SC} = 0$

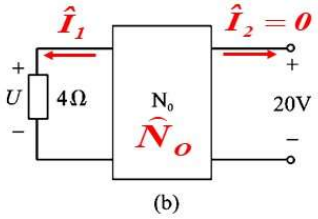
$$I_{SC} = \frac{U_S}{2} \qquad R_{eq} = \frac{U_{OC}}{I_{SC}} = 2\Omega$$

3、当R_L=2Ω时,有最大功率　　$U_S = (2 + R_L)\times I_L = 8V$

4-7、如图 (a)和(b)中的N_0线性无源网络，试求 (b)中电阻上的电压U。

解　**由特勒根定理：**





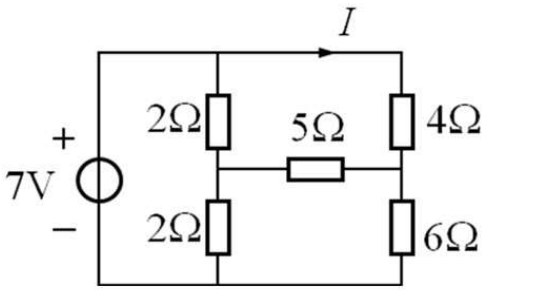
$$U_1\hat{I}_1 + U_2\hat{I}_2 + = \hat{U}_1 I_1 + \hat{U}_2 I_2$$

$$\hat{I}_1 = \frac{\hat{U}_1}{4} = \frac{U}{4}$$
$$\hat{U}_1 = U$$
$$U_1 = 20$$
$$\hat{U}_2 = 20$$

$$20\times\frac{U}{4} + 0\times\hat{I}_2 = U\times(-5) + 20\times 1$$

$$\boxed{U = 2V}$$

4-8、 如图电路，用互易定理求I=?



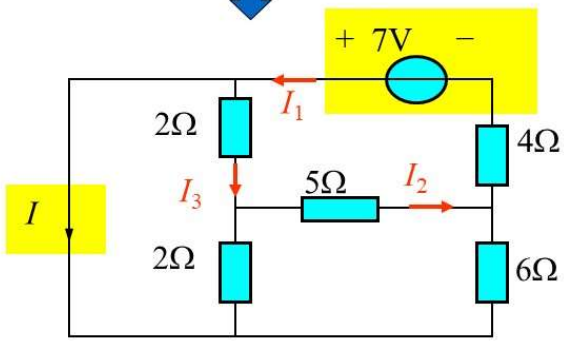
解 互易后，7V电源外接总电阻R:

I_1= 7/[(2//2+5)//6+4]=1A

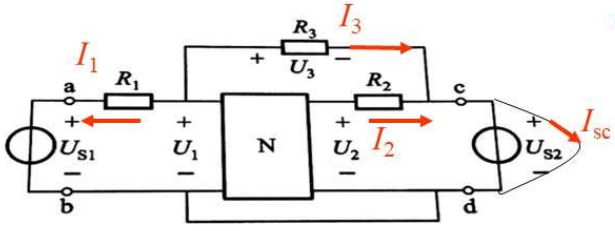
I_2 = 0.5 I_1=0.5A

I_3 = 0.5 I_2=0.25A

I= I_1−I_3 = 0.75A



4-9 求如图4-9所示电路中，N是无源线性非时变电阻电路，其中R_1=1Ω，R_2=2 Ω，R_3=3 Ω，U_{s1}=18V，U_{s2}=27V。当U_{s1}作用而U_{s2}用短路代替时，测得U_1=9V，U_2=4V。试求当共同作用时的U_3

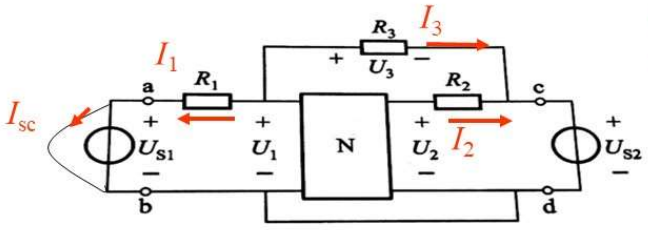


(1) 当U_{s1}作用而U_{s2}用短路代替

$$U_1^{(1)} = 9V, U_2^{(1)} = 4V$$

$$U_3^{(1)} = 9V, i_3^{(1)} = 3\,\mathrm{A}$$

$$i_2^{(1)} = 2A, I_{sc}^{(1)} = 5A$$



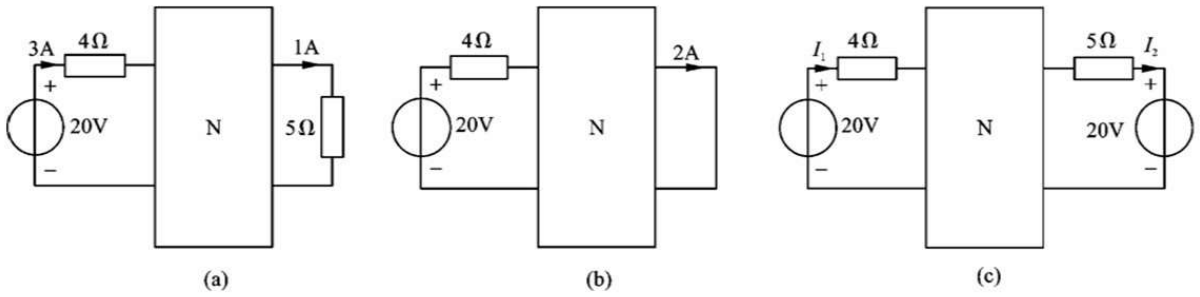
(2) 把U_{s1}换到c-d端口并提高到27V，U_{s1}用短路代替

$$I_1^{(2)} = I_{sc}^{(2)} = 5 * \frac{27}{18} = 7.5A$$

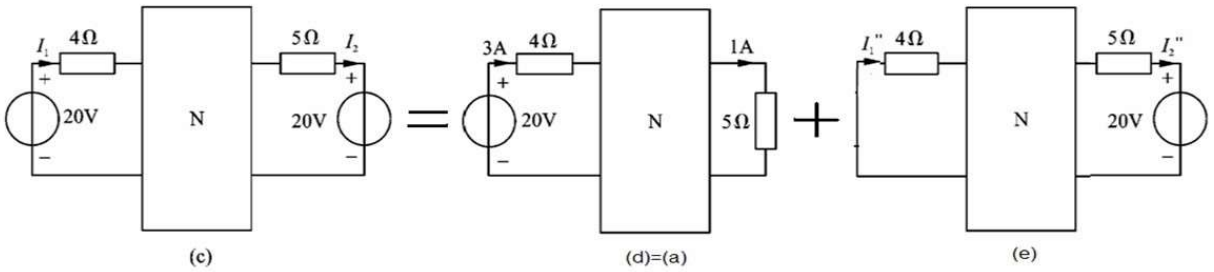
$$U_3^{(2)} = 7.5 - 27 = -19.5V,$$

$$U_3 = 9 - 19.5 = -10.5V$$

4-10 、试用互易定理求图 (c)所示电路中的电流I_1和I_2.



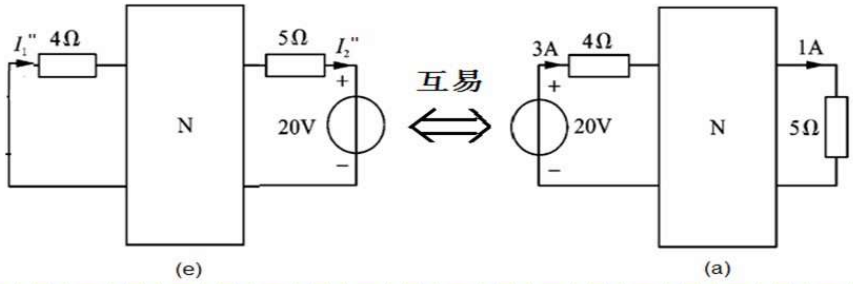
解 (1)对图(c)使用叠加得分图(d)、(e)，图(d)即图(a)，所以 $I_1^{'} = 3\,A, I_2^{'} = 1\,A$



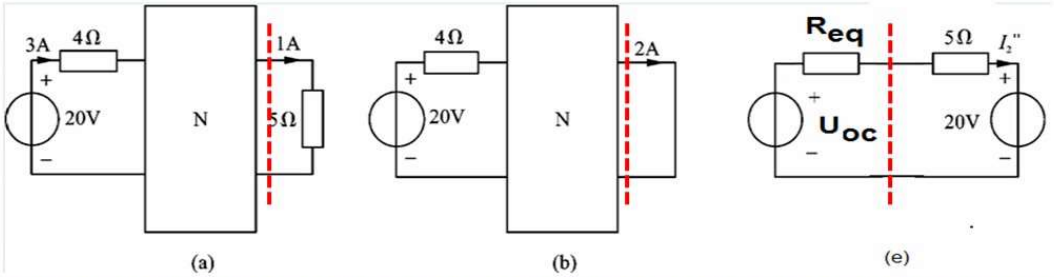
(2)对图(e)、图(a)使用互易，所以

$I_1^{''} = -1\,A$

注意方向

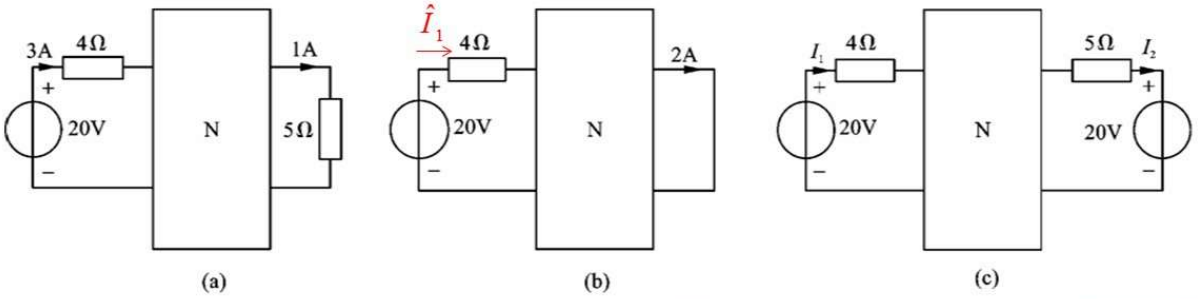


(3)对图(a)、图(b)、图(e)使用戴维南，得 $U_{oc} = 10\,V, R_{eq} = 5\,\Omega$



$\therefore I_2 = -1\,\mathrm{A}$

(4)各电流代数和 $I_1 = I_1^{'} + I_1^{''} = 2\,\mathrm{A}, \quad I_2 = -1\,\mathrm{A}$



解法二：求I_2

对图(a)和(b)应用特勒根定理2,可得

$$20\times\hat{I}_1+1\times5\times2=(-3)\times20+1\times0$$

$$\hat{I}_1=\frac{-70}{20}=-3.5\ \text{A}$$

再对图(b)和(c)应用特勒根定理2,并把前面求得的$I_1=2$ A和$\hat{I}_1=-3.5$ A代入,有

$$20\times\hat{I}_1+(5I_2+20)\times2=20\times(-I_1)+0\times I_2$$

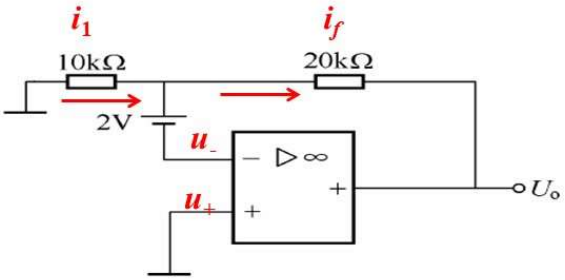
即

$$20\times(-3.5)+10I_2+40=20\times(-2)$$

故解得

$$I_2=\frac{-40-40+70}{10}=-1\ \text{A}$$

5-1 由理想运放构成的某负反馈电路如图所示，求电路的输出电压U_O。

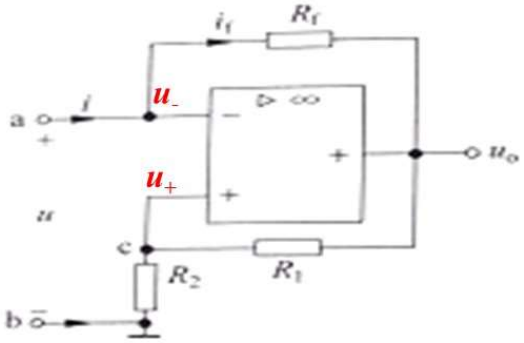


$$u_- = u_+ = 0$$

$$i_1 = -\frac{2}{10k} = i_f$$

$$U_o = -R_f i_f + 2 = 6\text{V}$$

5-2 求ab端口的电阻。



$$u_{ab} = u_+ = \frac{u_o}{R_1 + R_2} R_2$$

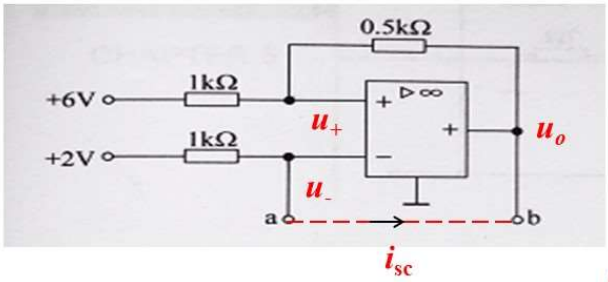
$$u_- - u_o = R_f i_f = R_f i$$

$$u_{ab} = u_+ = -\frac{R_2}{R_1} R_f i$$

$$R_{ab} = \frac{u_{ab}}{i} = -\frac{R_2}{R_1} R_f$$

5-3 求图所示电路两点之间的戴维南电路。

(1) 求u_{oc}



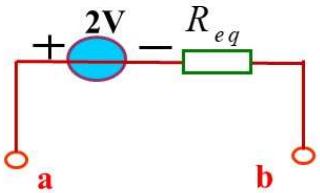
$$i_- = i_+ = 0, \quad u_- = u_+ = 2\text{V}$$

$$\frac{6 - u_+}{1k} = \frac{u_+ - u_o}{0.5k}$$

$$u_o = 1.5u_+ - 3 = 0V$$

$$u_{ab} = u_- - u_o = 2V$$

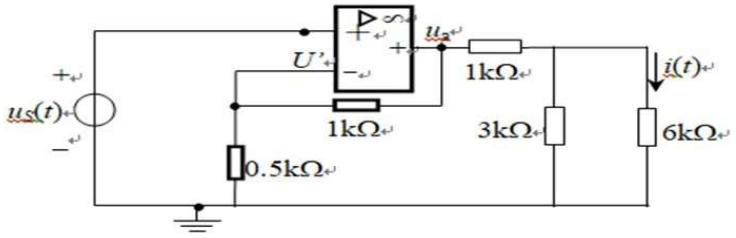
(2) 求i_{sc}



$$\left.\begin{array}{l} u_- = u_o = u_+ \\ \dfrac{6 - u_+}{1k} = \dfrac{u_+ - u_o}{0.5k} \end{array}\right\} u_o = u_+ = 6V$$

$$i_{sc} = \frac{2 - u_o}{1k} = -4mA \qquad R_{eq} = \frac{2}{-4m} = -500\Omega$$

5-4 已知题图所示电路中，电压源$u_S(t)=\sin 4t$ V，电阻$R_2=2R_1=1\text{k}\Omega$。求电流$i(t)$。



$$u_S=\frac{0.5}{1+0.5}u_a=\frac{1}{3}u_a$$

$$\Rightarrow u_a=3u_S$$

$$i=\frac{u_a}{1+3//6}\times\frac{3}{3+6}$$

$$=\frac{1}{9}u_a=\frac{1}{3}u_S$$

$$\Rightarrow i=0.333\sin 4t \text{ mA}$$